The Linux Command Line

From First Principles to Mastery

Calamity Paper QA

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How to use this book

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CHAPTER 1

Listing a directory: `ls` from the inode up

You have typed `ls` thousands of times. By the end of this chapter you will know what it asks the kernel for, why `ls -l` is slower than plain `ls`, and why it hangs on a directory with two million files.

> **Origin story — 1971, Unix V1**
>
> The first Unix manual already listed `ls`. Its job was small: read a directory and print the names inside it. Everything it has grown since then is an answer to one question: *what else might you want to know about those names?*

1.1 A directory is a file of names

> **In plain words**
>
> A directory is a list. Each line of the list holds a name and a number. The number points to the real file.

Definition: Directory entry (dentry)

A record inside a directory that pairs a file name with an *inode number*. The inode, not the name, holds the file's metadata and the location of its data.

Figure 1.1 shows the three layers. Plain `ls` only needs the left box; `ls -l` must visit the middle box once per entry.

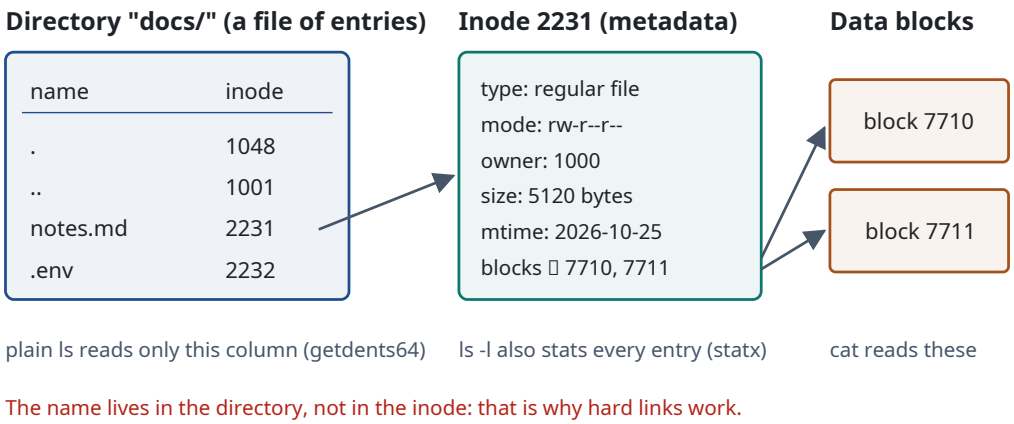


Figure 1.1: A directory maps names to inode numbers; the inode holds metadata and points to data blocks. Notice which layer each command has to touch.

Under the hood

Run `strace -e trace=getdents64,statx ls -l` on a small directory. You will see one `getdents64` call that returns all the names, then one `statx` call per file. That second loop is the cost of `-l`.

```
$ strace -c -e trace=getdents64,statx ls -l /etc > /dev/null
% time     seconds  usecs/call     calls    errors syscall
------ ----------- ----------- --------- --------- ----------------
 91.40    0.000851           3       241           statx
  8.60    0.000080          27         3           getdents64
```

1.2 The flags, grouped by what they change

Table 1.1 groups the flags you will meet most. The full list is in the GNU manual [1].

Table 1.1: Common `ls` flags, grouped by purpose (GNU coreutils).

Group	Flag	What it does
What to show	`-a`	Include entries starting with a dot, including `.` and `..`
	`-A`	Like `-a` but leaves out `.` and `..`
Detail	`-l`	Long format: type, mode, links, owner, size, time
	`-i`	Print each entry's inode number
Sizes	`-h`	Human-readable sizes, powers of 1024
	`--si`	Human-readable sizes, powers of 1000
Sorting	`-t`	Newest first, by modification time
	`-U`	Do not sort: directory order, fastest on huge directories

Gotcha — Sorting needs everything first

`ls` cannot print the first name until it has read every name, because sorting needs the whole list. On a directory with millions of entries that looks like a hang. `-U` skips the sort and streams.

Interview corner

"Why is `ls -l` slower than `ls` on a network filesystem?" A strong answer names the extra `statx` per entry and notes that each one may be a network round-trip.

Chapter summary

- A directory stores names and inode numbers; the inode stores everything else.
- `-l` costs one `statx` per entry; sorting costs reading every entry first.

Vocabulary

Inode
: The record holding a file's metadata and block locations.

Directory entry
: A name-to-inode pair stored inside a directory.

Exercises

Exercise 1.1 ★ Recall

What is the difference between `-a` and `-A`?

Exercise 1.2 ★ ★ ★ Interview

A colleague says `ls` "hangs" in `/var/spool/queue`. Explain why in terms of system calls, and give two commands that would list the directory without hanging.

Further reading

- The GNU `ls` manual [1]: read the sorting section closely.

APPENDIX A

Hints and solutions

Chapter 1

Solution to Exercise 1.1

`-a` lists every entry, including the special entries `.` and `..`; `-A` lists the hidden files but omits those two.

Solution to Exercise 1.2

`ls` must read every entry with `getdents64` before sorting, so nothing prints until the whole directory is read. Use `ls -U` to skip sorting, or `find . -maxdepth 1`, which streams.

Bibliography

[1] Free Software Foundation. ls invocation — GNU coreutils manual. https://www.gnu.org/software/coreutils/manual/html_node/ls-invocation.html, 2026. Accessed 2026-10-25.